

Lab 11 – My Solutions

1: Exercise: Zone Configuration:

The screenshot shows two windows from an Oracle VM VirtualBox environment. The top window is 'DC1 [Running] - Oracle VM VirtualBox', displaying the 'DNS Manager' console. The left pane shows a tree view of DNS zones under 'DC1', including 'Forward Lookup Zones' and 'Reverse Lookup Zones'. The right pane displays a list of DNS records:

Name	Type	Data
www	Alias (CNAME)	pc200.apr.internal.
pc200	Host (A)	192.168.56.200
lab		
dev		
dc1	Host (A)	192.168.56.100
(same as parent folder)	Start of Authority (SOA)	[8], dc1.contoso.internal, ...
(same as parent folder)	Name Server (NS)	dc1.contoso.internal.

The bottom window is 'CL1 [Running] - Oracle VM VirtualBox', showing a PowerShell console window titled 'Administrator: PowerShell'. It displays the results of two 'Test-Connection' commands:

```
PS C:\Users\Administrator.CONTOSO> Test-Connection www.apr.internal

Destination: www.apr.internal

Ping Source      Address           Latency  BufferSize Status
-----
1 c11            192.168.56.200    0        32 Success
2 c11            192.168.56.200    1        32 Success
3 c11            192.168.56.200    1        32 Success
4 c11            192.168.56.200    1        32 Success

PS C:\Users\Administrator.CONTOSO> Test-Connection pc200.apr.internal

Destination: pc200.apr.internal

Ping Source      Address           Latency  BufferSize Status
-----
1 c11            192.168.56.200    0        32 Success
2 c11            192.168.56.200    0        32 Success
3 c11            192.168.56.200    0        32 Success
4 c11            192.168.56.200    0        32 Success

PS C:\Users\Administrator.CONTOSO>
```

DC1 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

DNS Manager

FileActionViewHelp

DNS

DC1

Forward Lookup Zones

msdcs.contoso.internal

apr.internal

dev

lab

contoso.internal

lab.apr.internal

Reverse Lookup Zones

Trust Points

Conditional Forwarders

Name	Type	Data	Timestamp
pcdev	Host (A)	192.168.56.200	static
wwwdev	Alias (CNAME)	pcdev.dev.apr.internal.	static

Search

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CL1 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

PS C:\Users\Administrator.CONTOSO> Test-Connection pcdev.dev.apr.internal

Destination: pcdev.dev.apr.internal

Ping Source	Address	Latency (ms)	BufferSize (B)	Status
1 cl1	192.168.56.200	0	32	Success
2 cl1	192.168.56.200	1	32	Success
3 cl1	192.168.56.200	1	32	Success
4 cl1	192.168.56.200	1	32	Success

PS C:\Users\Administrator.CONTOSO>

2: Exercise: Zone Delegation:

The image shows two windows from an Oracle VM VirtualBox environment. The top window is the DNS Manager for a server named DC1. The left pane shows a tree view of DNS zones, with 'lab.apr.internal' selected under the 'apr.internal' zone. The right pane displays a list of DNS records for the selected zone:

Name	Type	Data
www	Alias (CNAME)	pc200.apr.internal.
pc200	Host (A)	192.168.56.200
lab	Host (A)	192.168.56.100
dev	Host (A)	192.168.56.100
dc1	Host (A)	192.168.56.100
(same as parent folder)	Start of Authority (SOA)	[8], dc1.contoso.internal., ...
(same as parent folder)	Name Server (NS)	dc1.contoso.internal.

The bottom window is a PowerShell terminal running as Administrator. It shows the execution of 'ipconfig /flushdns' and 'Test-Connection l200.lab.apr.internal', which successfully flushes the DNS resolver cache and performs a ping test to the specified destination.

```
PS C:\Users\Administrator.CONTOSO> ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.
PS C:\Users\Administrator.CONTOSO> Test-Connection l200.lab.apr.internal

Destination: l200.lab.apr.internal

Ping Source      Address          Latency BufferSize Status
-----
1 cl1            192.168.56.200  0      32 Success
2 cl1            192.168.56.200  1      32 Success
3 cl1            192.168.56.200  1      32 Success
4 cl1            192.168.56.200  1      32 Success

PS C:\Users\Administrator.CONTOSO> |
```

3: Exercise: The hosts File:

The image displays two windows side-by-side. The left window is an Administrator PowerShell terminal showing the results of a `Test-Connection` command. The right window is a File Explorer showing the contents of the `etc` directory, with the `hosts` file open in a text editor.

PowerShell Test Results:

```
PS C:\Users\Administrator.CONTOSO> Test-Connection www.apr.internal

Destination: www.apr.internal

Ping Source      Address          Latency BufferSize Sta
-----
1 cll            192.168.56.200  0      32 Suc
2 cll            192.168.56.200  0      32 Suc
3 cll            192.168.56.200  1      32 Suc
4 cll            192.168.56.200  0      32 Suc

PS C:\Users\Administrator.CONTOSO> Test-Connection www.apr.internal

Destination: www.apr.internal

Ping Source      Address          Latency BufferSize Sta
-----
1 cll            8.8.8.8          7      32 Suc
2 cll            8.8.8.8          7      32 Suc
3 cll            8.8.8.8          8      32 Suc
4 cll            8.8.8.8          7      32 Suc

PS C:\Users\Administrator.CONTOSO> |
```

Hosts File Content:

```
#
# For example:
#
# 102.54.94.97  rhino.acme.com    # source server
# 38.25.63.10  x.acme.com           # x client host

# localhost name resolution is handled within DNS itself.
# 127.0.0.1    localhost
# ::1          localhost
8.8.8.8 www.apr.internal
```

4: Exercise: DNS and Active Directory:

Navigate to: **DNS console → DC1 → _msdcs.contoso.internal**

Record path	What it contains	Meaning
_tcp.dc._msdcs.contoso.internal	SRV records for all Domain Controllers	Clients use this to find <i>any</i> DC for Kerberos/LDAP authentication
_tcp.pdc._msdcs.contoso.internal	SRV record for the PDC Emulator	Points to the single DC holding the PDC Emulator FSMO role (handles password changes, time sync, legacy NT clients)

The entire _msdcs zone is auto-populated by AD DS — it's how clients, DCs, and replication all discover each other without manual configuration. Every time a DC starts, it re-registers its SRV records here.

